

Date

Group Members

Nighat Naseem

53740

• Uniform violation

Anosha Perwaiz

56200

• Uniform violation

Afifa Amna

55466

• Uniform violation

Nimra Tariq

54909

Simab Malik

54910

• Uniform violation

Areeza Afridi

55254

• Uniform violation

Artificial Intelligence

Assignment # 01

Submitted to Ma'am Shumaila

[Background :-] x

- In our daily life people waste 10 to 15 minutes daily deciding what to wear.
- The common problem occurs choosing wrong outfit for weather, mismatch occasion and oftenly repeating cloths.
- People also may forget whats in the closet.

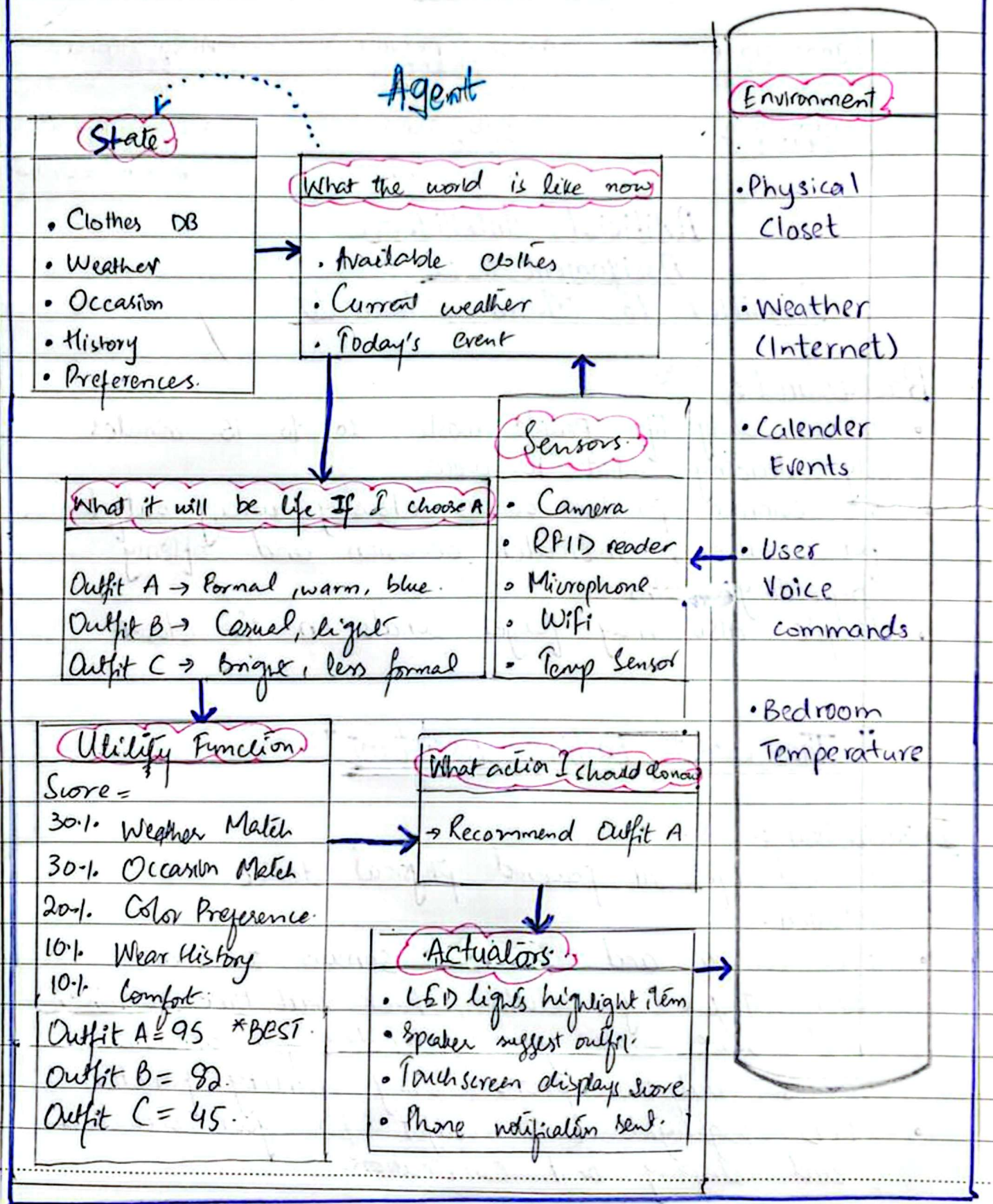
" The Closet Fashion Assistant "

[Introduction :-] x

- A real life AI powered physical device installed.
- use camera and RF-ID sensors to scan and track your cloths, check real time weather and sync with your calenders for occasions. Scores outfit options, using utility function
- LED highlights best outfit for your occasions and display best combination.

Date

Architecture of Agent (Utility - Based)



* The PEAS:-

Component	Description
• Performance	user satisfaction, outfit appropriateness time saved, user likes/dislikes saved outfit.
• Environment	physical closet, real clothes with RFID tags, bedroom, weather outside, calendar events.
• Actuators	LED lights (on clothes), speakers (voice output), touch screen display
• Sensors	Camera, microphones, RFID reader wifi modules, temperature sensors . calendar sync.

* Agent Type :- X

⇒ Utility - Based Agent (with learning capability)

Reason:

It doesn't just find any outfit - it scores multiple outfits and selects the one with highest utility value (best score) based on weather, occasion, color, ~~highest~~ history, and comfort.

* Agent function :- X

$f(\text{percept}) \rightarrow \text{action}$.

Maps current situation (weather + occasion + available clothes + user history) Best outfit suggestion with highest utility score.

Date

Percepts	Sensors	Actions	Actuators
→ Weather (rainy/sunny/cold)	WiFi module/ ✓ temp sensor	Suggest outfit with LED	LED lights
→ Occasion (formal/casual/party)	Microphone/ ✓ calendar sync	Suggest outfit with LED	LED lights
→ Available clothes	camera/RFID ✓	Suggest outfit	LED lights
→ Clean/dirty status	RFID reader	Voice announcement	speaker
→ last worn date	Database query	show outfit ^{on} screen	touch screen
→ User preferences	history/Database	send to phone	notification
→ user feedback ("like this")	touch screen	save fav. outfit	Database server

so Percept Sequence: X

→ Time to (Morning 8:00 AM):

percept →

weather : rainy, 18°C.

occasion = job interview

available = blue-shirt (clean), white-shirt (clean), black-pants (clean),
jeans (dirty).

last-worn : blue-shirt (3 days ago), black-pants (5 days ago)

user-mood = in a hurry.

Time (After suggestions):

percept →

user-feedback = selected outfit B instead of A. ✓

new-state = gray-shirt now being worn.